CY4001: Industrial Chemistry

Unit 3:Fuel Chemistry
(Liquid Fuels)

William Merriam Burton (1865 – 1954)

- William Meriam Burton was a Chemist and oil industry executive.

- Burton's thermal cracking process, patented in 1913, used high temperatures at high pressures to "crack," or break down, heavier hydrocarbons into lighter molecules.

- Burton demonstrated the value of laboratory research and testing

- The cracking process he developed more than doubled the potential yield of gasoline from crude oil. During its first 15 years in use the process saved more than one billion barrels of crude oil.

Burton started work at Standard Oil in Cleveland as a chemist. There he later served as assistant superintendent and in 1895 became superintendent of the refinery. He was elected a director of the company in 1911, vice president in 1915, and president in 1918.

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Liquid Fuels

- Liquid fuels are classified into two categories:
  (a) natural or crude oil fuels
  (b) manufactured fuels.

Advantages of Liquid Fuels

- Possess higher calorific value per unit mass than solid fuels.
- Burn without dust, ash, clinkers, etc.
- Firing is easier and also fire can be extinguished easily by stopping liquid fuel supply.
- Easy to transport through pipes.
- Stored indefinitely without any loss.
- Clean in use and economic to handle.
- Loss of heat in chimney is very low due to greater cleanliness.
- Require less excess air for complete combustion.
- Require less furnace space for combustion.

Disadvantages of Liquid Fuels

- The cost of liquid fuel is relatively much higher as compared to solid fuel.
- Costly special storage tanks are required for storing liquid fuels.
- There is a greater risk of fire hazards, particularly, in case of highly inflammable and volatile liquid fuels.
- They give bad odour.
- For efficient burning of liquid fuels, specially constructed burners and spraying apparatus are required.

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Petroleum and Its Characteristics

- Petroleum, or crude oil is a naturally occurring oily, bituminous liquid composed of various organic chemicals.
- It is found in large quantities below the surface of the earth and is used as a fuel and as a raw material in the chemical industry.
- In addition, petroleum and its derivatives are used in the manufacture of medicines, fertilizers, foodstuffs, plastics, building materials, paints, etc.

The approximate composition of petroleum is:

C = 80 – 84%

H = 10 – 14 %

S = 0.1 – 0.5 %

N + O = Negligible

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Petroleum and Its Characteristics

Petroleum's are graded according to the following physicochemical properties:

- Specific gravity & Calorific value
- Flash point or ignition point
- Viscosity & Sulphur contents
- Moisture and sediment content
- Specific heat and coefficient of expansion

Classification of Petroleum:

- The chemical nature of crude petroleum varies with the part of the world in which it is found.
- Three broad classes of crude petroleum exist:
  (a) the paraffin types
  (b) the asphaltic types
  (c) the mixed-base types.

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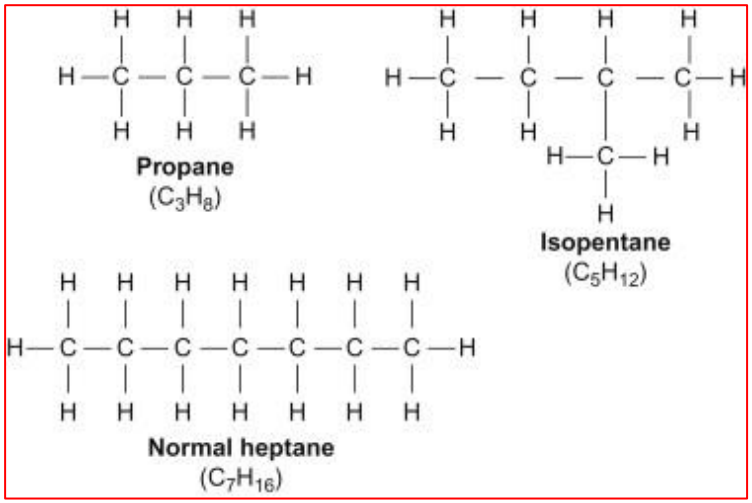
Classification of Petroleum:

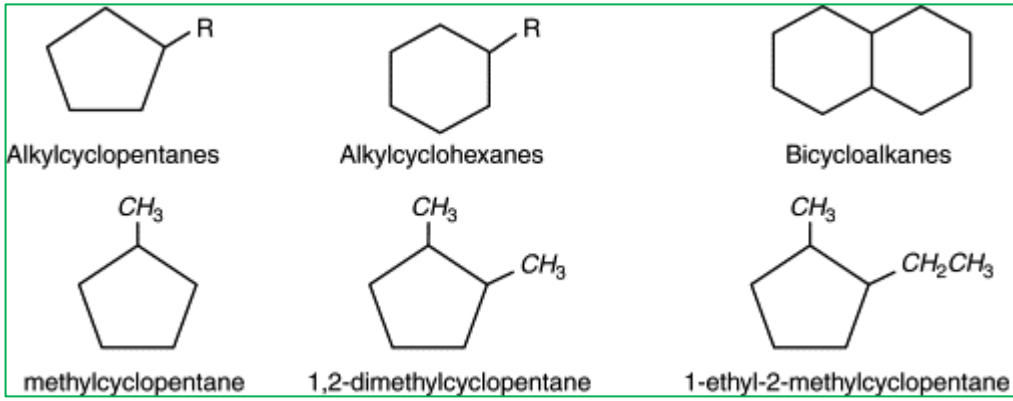
(a) The paraffin types:

- This type of petroleum is mainly composed of the paraffins or saturated hydrocarbons (C_nH_{2n+2}) from CH_4 to $C_{35}H_{72}$ and a little of the naphthenes and aromatics.
- The number of hydrogen atoms is always two more than twice the number of carbon atoms.
- The hydrocarbons from $C_{18}H_{38}$ to $C_{35}H_{72}$ are sometimes called waxes.



(b) The asphaltic types:

- It mainly contains cycloparaffins or naphthenes with smaller amount of paraffins and aromatic hydrocarbons.



(c) Mixed base types:

- It contains both paraffinic and asphaltic hydrocarbons and is generally rich in semi-solid waxes.

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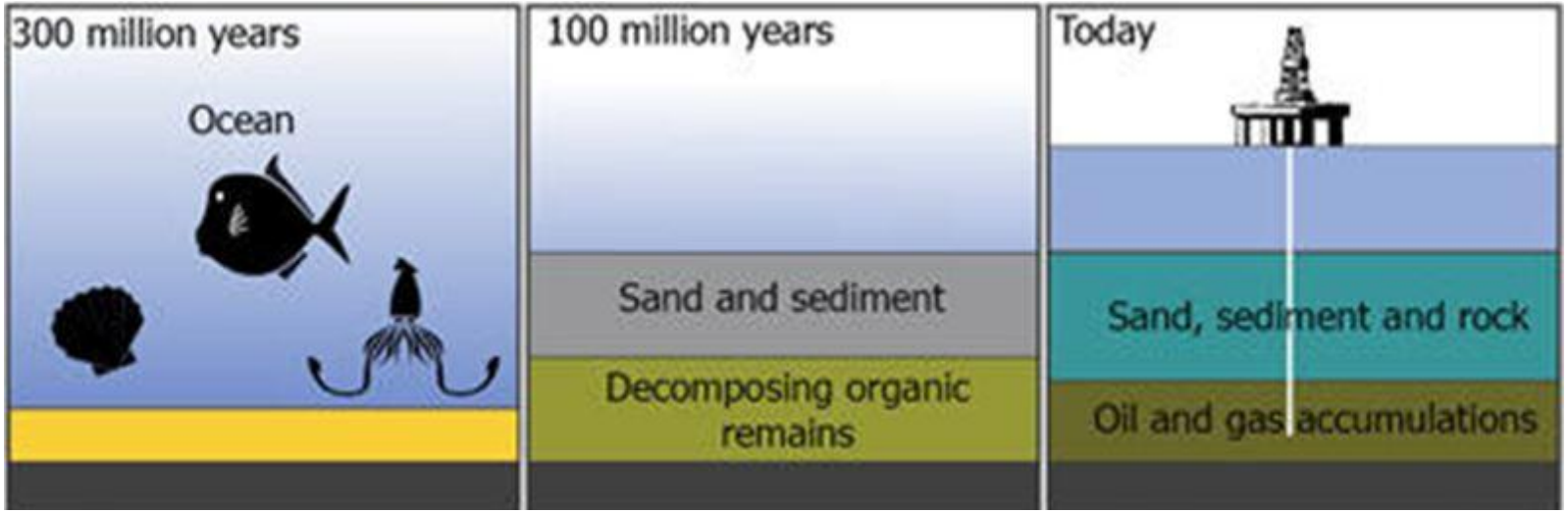
Formation of Petroleum

- Petroleum is formed under the earth's surface by the decomposition of marine organisms.
- The remains of tiny organisms that live in the sea and, to a lesser extent, those of land organisms are carried down to the sea by rivers, and of plants that grow on the ocean bottoms are enmeshed with the fine sands and silts that settle to the bottom in quiet sea basins.
- Such deposits, which are rich in organic material, become the source for the generation of crude oil.



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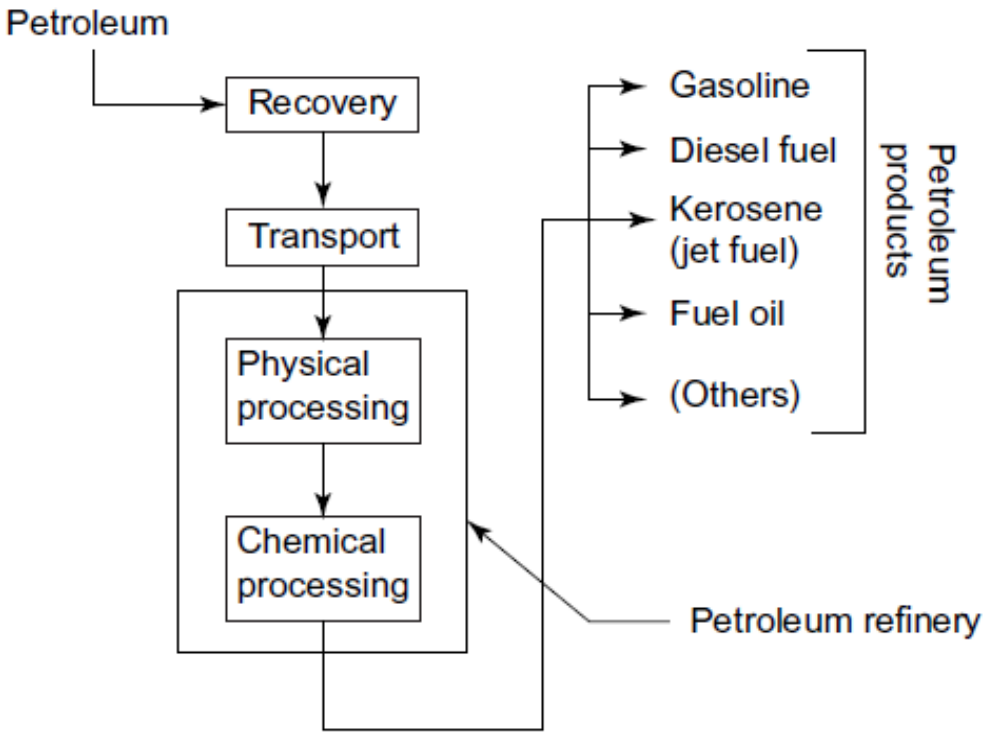
Petroleum Utilization and Refining:

- Petroleum utilization is a much more complex process than coal utilization.
- The preparation of petroleum before being sold to consumers is very extensive.
- The reason is the chemical structure of different crude oils may be very different.
- Furthermore, a large number of different products are obtained from the petroleum refinery.

Pathway to petroleum utilization



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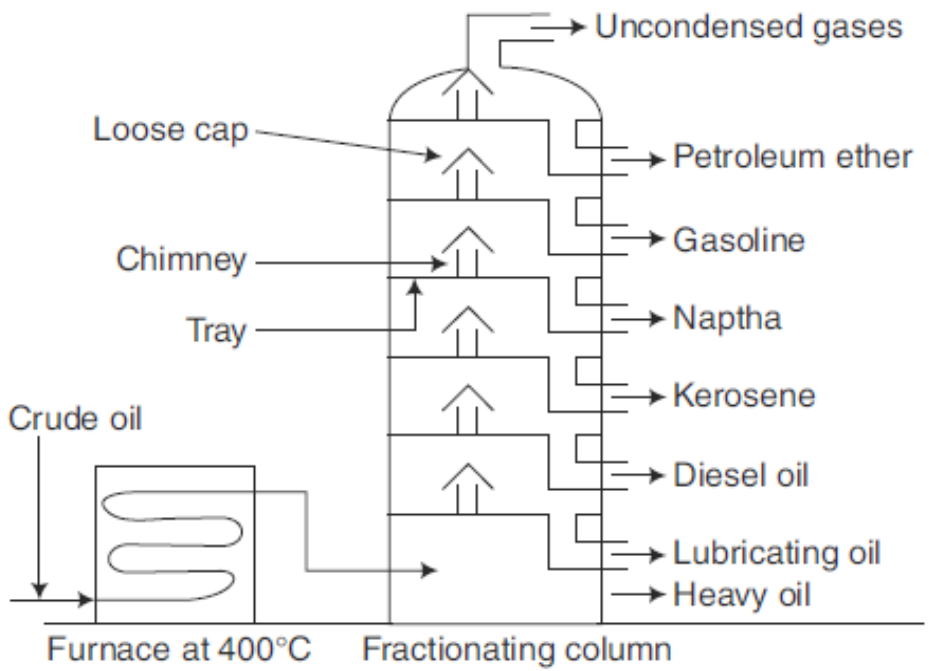
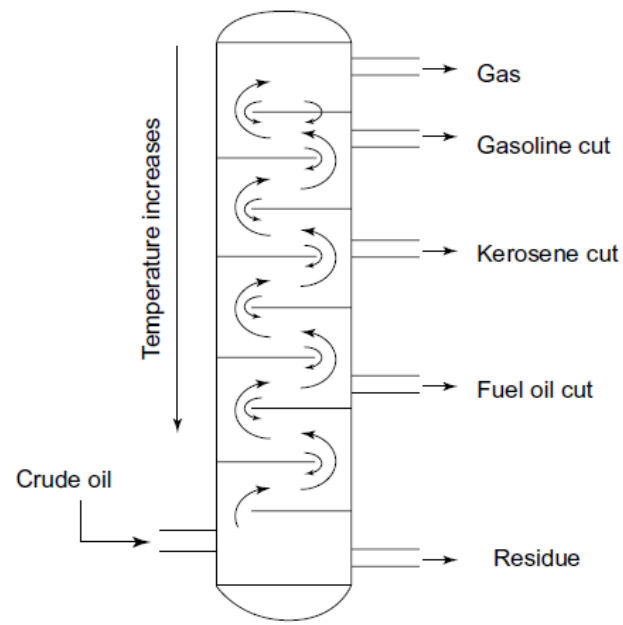
Petroleum Refining:

- The process of removing impurities and separating the oil into various fractions having different boiling points is known as the refining of petroleum.
- Once oil has been produced from an oil field, it is treated with chemicals and heat to remove water and solids, and the natural gas is separated.
- The oil is then stored in a tank, and later transported to a refinery by truck, railroad tank car, barge, or pipeline.

Distillation of various fractions from petroleum





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Various fractions of petroleum distillation

S. No	Name of the Fraction	Hydrocarbons in terms of Carbon Atom	Boiling Point in °C	Uses
1.	Uncondensed gases	C_1–C_4	< 30	LPG—domestic fuel
2.	Petroleum ether	C_5–C_7	30–70	Solvents for varnish and rubber
3	Gasoline or petrol	C_5–C_8	40–120	Fuel in ICE of automobiles and aeroplanes
4	Naphtha or Solvent spirit	C_9–C_{10}	120–180	Solvent in dry cleaning
5	Kerosine oil	C_{10}–C_{16}	180–250	Domestic fuel, Jet engine fuel.
6	Diesel oil or gas oil	C_{15}–C_{18}	250–320	Diesel engine fuel
7	Heavy oil	C_{17}–C_{30}	320–400	To produce cracked petrol

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Petrol or Gasoline

- It is a low boiling fraction of petroleum obtained between 40-120°C.
- It is a mixture of hydrocarbons pentane to nonane (in terms of carbon atoms C5–C9).
- Its calorific value is about 11, 250 kcal/kg.
- It is used as fuel in the Internal combustion engine (ICE) of automobiles and airplanes.

Knocking of Petrol

Knocking is a phenomenon that occurs in gasoline. Sharp sounds caused by premature combustion of part of the compressed air-fuel mixture in the cylinder.. It's essentially combustion at the wrong time.

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Knocking and Octane ratings

What is knocking?

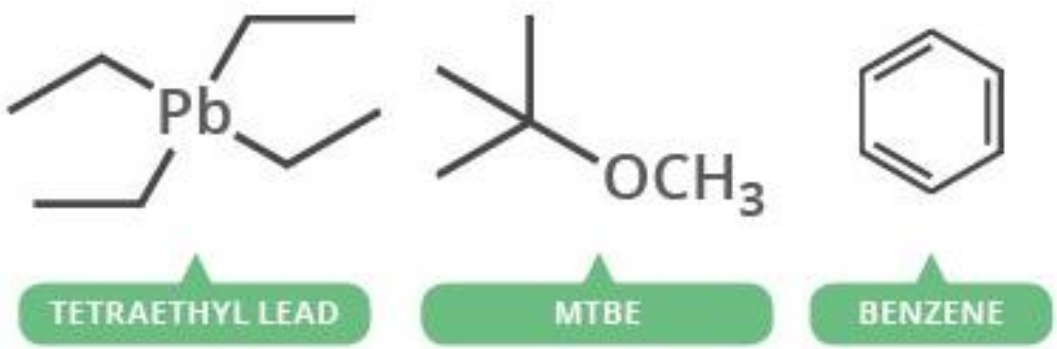
Knocking occurs when fuel's combustion doesn't occur in sync with the engine cycle. It lowers the engine efficiency and eventually damage the engine.

Octane ratings

Octane ratings gauge how well fuel avoids this problem. Higher octane number indicates lesser knocking.

Anti-knocking

- Compounds can be added to petrol to increase its octane rating. Tetraethyl-lead is one of the anti-knocking agents, but it is banned in most countries as it releases toxic lead fumes.
- Alternative anti-knocking agents used in unleaded petrol include methyl tertiary butyl ether (MTBE), ethanol, benzene and toluene.



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Octane Number or Octane Rate:

In the early days of the automotive industry, there was no way to tell if gasoline would knock in a new engine, or if a new gasoline would knock in any engine. The only way to find out if it would knock would be to fill up your tank and crank the engine. This problem was solved in 1926 by **Dr. Graham Edgar** of the Ethyl Gasoline Corporation, then a division of General Motors and Standard Oil. Dr. Edgar devised the octane scale by testing the knocking tendency of every compound that was similar to gasoline. He found that isooctane would not knock in any engine under any operating conditions, while n-heptane would always knock in any engine. By mixing isooctane and n-heptane in different amounts, he obtained fuels of all qualities, the percentage of isooctane in the mixture being the *octane number*. Commercial gasoline was compared to these mixtures in test engines to determine their octane number. When the scale was developed, commercial gasoline had octane ratings between 40 and 75, and the best were brought up to a rating of 87 by the addition of tetraethyl lead, making leaded gasoline. Today, due to environmental reasons, tetraethyl lead has been outlawed in the United States and many other parts of the world. However, we do not really need it, because new and better processes allow oil companies to make gasoline for our automobiles with octane ratings as high as 94.

From Frank A. Howard, *Buna Rubber: The Birth of an Industry.* New York: D. van Nostrand Company, Inc., 1947.

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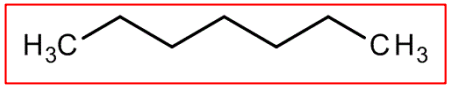
Octane Number or Octane Rate:

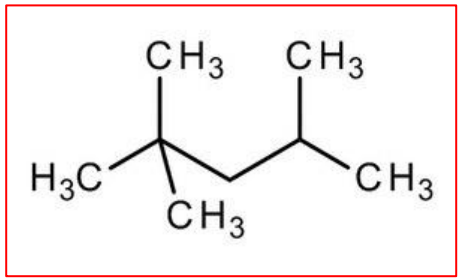
- A figure indicating the anti-knock properties of a fuel, based on a comparison with a mixture of isooctane and heptane.

- Greater the octane number, greater is the resistance to knocking.

- Among alkanes, n-heptane knocks severely, while iso-octane has a high resistance to knocking.



n-heptane (octane number = 0)



iso-octane (octane number =100)

- For the scale proposed to indicate the anti-knock properties of gasoline, n-heptane was arbitrarily assigned an octane number of zero and iso-octane was arbitrarily assigned a value of 100. By blending these two hydrocarbons in various proportions, primary reference fuels were prepared.

इंडियनऑयल
IndianOil

XP100
100 octane premium petrol

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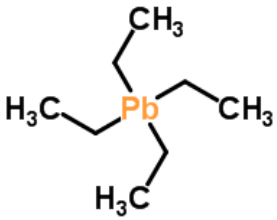
Anti-Knocking Agents:

- The octane rating of gasoline samples can be increased by the addition of certain organometallic compounds called anti-knocking agents and the process is called 'doping'.
- An extensively used anti-knocking agent is tetraethyl lead (TEL).



$\longrightarrow$ **Pb + PbO**

- It is believed that during combustion of gasoline, TEL forms Pb and PbO.
- These species act as free-radical chain inhibitors and thus curtail the propagation of the explosive chain reaction and thereby minimizing knocking.
- If TEL alone is used, the species Pb and PbO may get deposited on engine parts and cause mechanical damage. The vapors of Pb and PbO may pollute the air.
- TEL is always used along with ethylene dibromide or ethylene dichloride.
- The functions of these ethylene derivatives is to convert the less volatile Pb and PbO into more volatile $PbBr_2$ or $PbCl_2$ which escapes into air along with exhaust gases.

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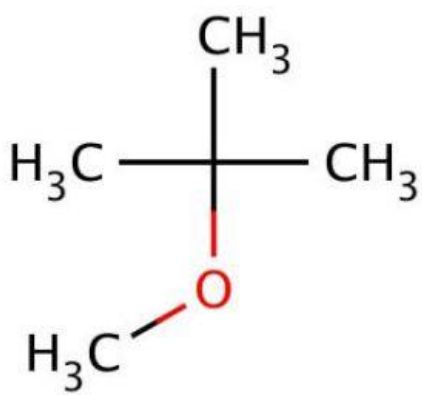
Unleaded Petrol:

- The petrol in which the enhancement of octane rating is accomplished without the addition of lead compounds is referred to as unleaded petrol.
- To improve its octane number, concentration of high octane components (like isopentane, isooctane, ethylbenzene, isopropyl benzene, etc.) is increased by the process of reforming.
- Compounds like methyl tertiary butyl ether (MTBE) can also be added to improve octane number of unleaded petrol.
- MTBE provides oxygen (of ether group) for combustion of petrol in IC engines, thereby reducing considerably the formation of peroxy compounds (which causes knocking).



MTBE

Advantages of Unleaded Petrol

- The harmful effects of discharge of poisonous lead and its compounds through the exhaust of automobiles is avoided.
- One of the major advantages of using unleaded petrol is that it permits the attachment of a catalytic converter to the exhaust pipe in automobiles.

Straight-run gasoline

- The crude oil on fractional distillation yields only about 15–20 % gasoline. This is known as straight-run gasoline.
- The quality of straight-run gasoline is not so good. It contains mainly straight chain paraffin, which ignites readily and more rapidly than any other hydrocarbons.
- Hence, it produces knocking (unwanted sound) in IC engines.
- To overcome these difficulties and to improve the quality of gasoline, long-chained, higher-boiling hydrocarbons are converted into shorter-chained, lower-boiling hydrocarbons - cracking
- The gasoline obtained by cracking is called cracked gasoline.

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Cracking

- The formation of low boiling hydrocarbons of low molecular weight by the decomposition (breaking of C-C and C-H bonds) of high boiling hydrocarbons of high molecular weight is called cracking.

$$\underset{\substack{n\text{-tetradecane} \\ \text{B.pt. } 253.7^\circ\text{C}}}{C_{14}H_{30}} \longrightarrow \underset{\substack{n\text{ pentane} \\ \text{B. pt : } 36^\circ\text{C}}}{C_5H_{12}} + \underset{\substack{\text{nonene} \\ \text{B.pt. } 143^\circ\text{C}}}{C_9H_{18}}$$

Cracking

- Thermal cracking (Carried out a high temperature and pressure in the absence of catalyst)
- Catalytic cracking (Carried out in presence of catalyst (Al_2O_3 + Sio_2) at of much lower temperature and pressure)

$Al_2O_3 + SiO_2$

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Thermal Cracking

- Thermal cracking involves subjecting hydrocarbons in crude oil to high heat and pressure to break the molecular bonds. This breaks down long-chained, higher-boiling hydrocarbons into shorter-chained, lower-boiling hydrocarbons

- The actual reaction is known as homolytic fission and produces alkenes, which are the basis for the economically important production of polymers

- Cracking may be carried out either in liquid phase or vapour phase.

  (a) Liquid phase thermal cracking (b) Vapour phase thermal cracking

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Mechanism of Thermal Cracking

Thermal cracking is believed to follow free radical mechanism. The mechanism can be illustrated by taking the example of n-nonane.

Initiation

$$CH_3(CH_2)_7CH_3 \xrightarrow{\text{heat}} CH_3(CH_2)_3CH_2 + \dot{C}H_2(\dot{C}H_2)_2CH_3$$

Propagation

The free radicals formed are thermally unstable and undergo fission at the β position to yield a new radical and an olefin.

$$CH_3-CH_2CH_2-CH_2-\dot{C}H_2 \rightarrow CH_3-CH_2-\dot{C}H_2 + CH_2=CH_2$$

This β scission continues, till a thermally more stable radical is formed.

$$CH_3-CH_2-\dot{C}H_2 \rightarrow \dot{C}H_3 + CH_2=CH_2$$

$$CH_3(CH_2)_7CH_3 + \dot{C}H_3 \rightarrow CH_3(CH_2)_5\dot{C}H-CH_2-CH_3 + CH_4$$

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Catalytic Cracking

- Carried out in presence of catalyst (Al_2O_3 + SiO_2) at of much lower temperature and pressure.

Advantages of Catalytic Cracking

- The octane number of gasoline produced is high.
- The yield of gasoline is also high.
- The process can be better controlled.
- The product contains a very little amount of undesirable sulphur.
- There is a saving in production costs since high temperatures and high pressures are not needed.
- In catalytic cracking, external fuel is not required. The necessary heat is obtained by burning off the coke deposited on the catalyst itself, during the regeneration process.
- The gasoline formed contains much less gum and gum forming compounds.

Catalytic Cracking Methods

- Fixed-bed catalytic cracking
- Fluidized (moving) bed catalytic cracking

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Reforming of Petrol:

- Reforming is a process designed to increase the volume of gasoline that can be produced from a barrel of crude oil.

- Hydrocarbons in the naphtha stream have roughly the same number of carbon atoms as those in gasoline, but their structure is generally more complex.

- Reforming rearranges naphtha hydrocarbons into gasoline molecules.

Catalytic Reforming:

- It is the process of upgrading gasoline (increasing its octane number) in presence of a catalyst.

- The increase in octane number occurs through structural modifications such as conversions of straight hydrocarbons into branched, cyclic and aromatic hydrocarbons.

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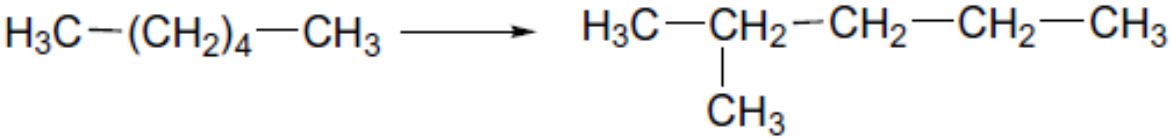
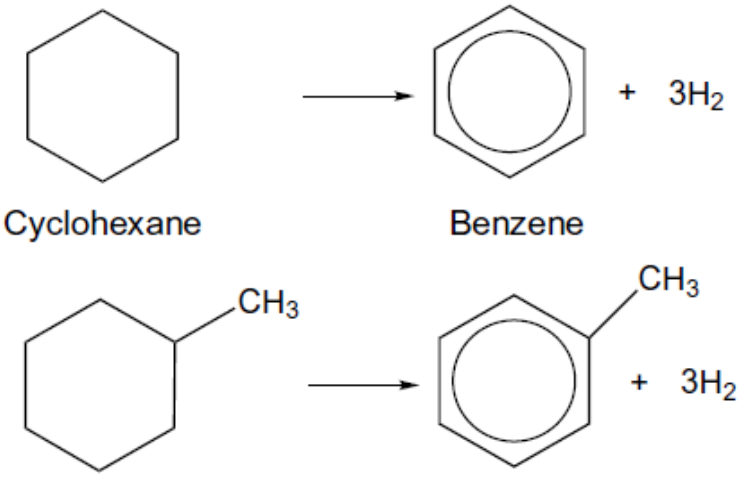
Reforming Reactions

a) Isomerization: The conversion of straight chain hydrocarbons into branched chain hydrocarbons.



n.hexane 2.methyl pentane

b) Dehydrogenation: Dehydrogenation of Cycloalkanes to produce aromatic compounds.



Cyclohexane Benzene

Methyl cyclohexane Toluene

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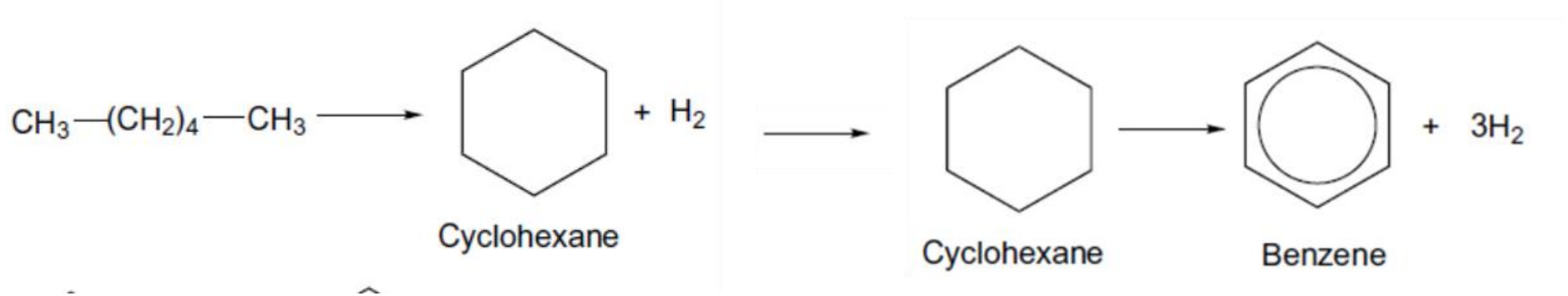
Reforming Reactions

c) Cyclization and dehydrogenation: **Cyclization of straight chain hydrocarbons followed by dehydrogenation to produce aromatic hydrocarbons.**



d) Hydro cracking: **Hydro cracking of n-Paraffins to produce light gases that are removed from gasoline fraction.**

$$CH_3—(CH_2)_8—CH_3 + H_2 \xrightarrow{Cat} 2CH_3—(CH_2)_3—CH_3$$

n-Decane n-Pentane

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Diesel

- Diesel is a product from crude oil in the distillation window between 250 °C and 320 °C.

- It consists of approximately 75% aliphatic hydrocarbons (C_{15} to C_{18}) and about 25% aromatic hydrocarbons (e.g. benzene, styrene).

- The basic difference between petrol engine and diesel engine is that, in case of a diesel engine, the fuel is ignited with the help of heat generated by the compression of air inside the combustion chamber and not by spark.

- The combustion of fuel in diesel engine is not instantaneous as that of petrol engine. There is a time lag between fuel injection and its ignition. It is called 'ignition delay' or 'ignition lag'.

Diesel engines are on the verge of extinction due to strict emission regulations.

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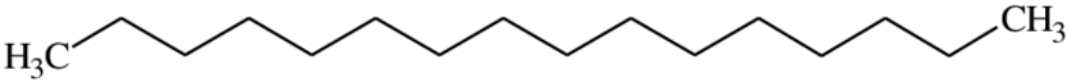
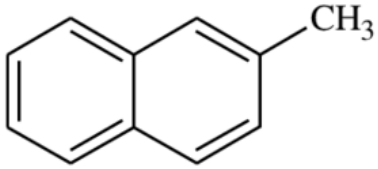
Cetane Number: (measurement of the quality of diesel fuel)

- Cetane number has been introduced to grade the diesel. It is a measure of ease at which fuel ignites under compression.

- n-Hexadecane or Cetane has very short ignition delay and arbitrarily its cetane number is assigned to be 100; whereas methylnaphthalene has lower ignition delay compared to other diesel fuels and it's cetane value has been assigned to be 0.



n-Hexadecane (cetane), cetane number = 100



1-Methylnaphthalene, cetane number = 0

The order of cetane number is as follows.

n-alkane > naphthalene > alkenes > branched alkanes > aromatics

Cetane no. can be increased by adding "Pre ignition drops" like Ethyl-nitrate, Iso-amyl alcohol, octane peroxide, etc.

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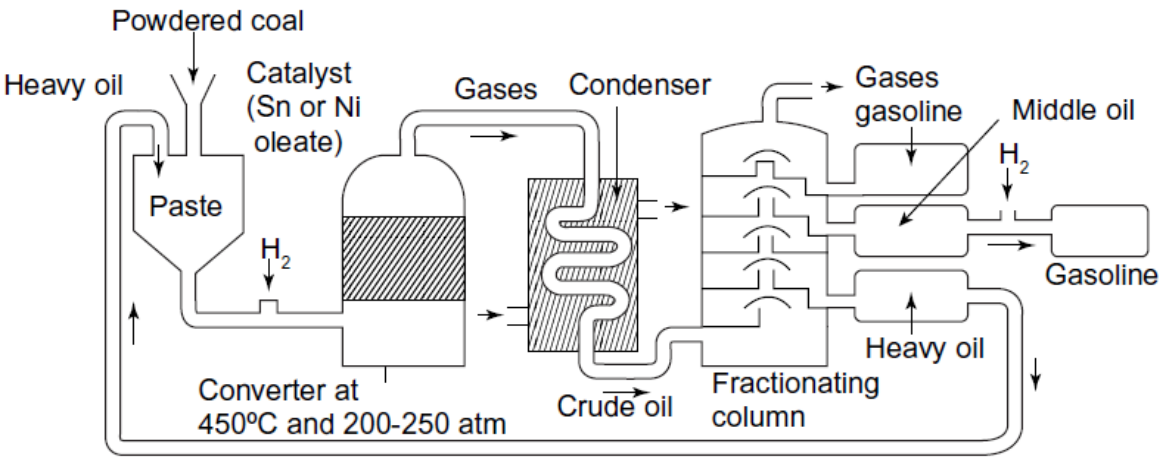
Synthetic Petrol:

- Synthetic petrol is a mixture of alkanes with composition resembling that of petrol, obtained artificially from coal. The two important methods for preparation are: Bergius process and Fischer-Tropsch process.

Bergius process:

The Bergius process is a method of production of liquid hydrocarbons for use as synthetic fuel by hydrogenation of high-volatile bituminous coal at high temperature and pressure.



$$\text{Coal dust} + H_2 \xrightarrow[\text{(450°C, 200 atm)}]{\text{Catalyst}} \text{Mixture of hydrocarbon} \xrightarrow[2\ \text{(Cracking)}]{1(H_2)} \text{Crude oil}$$

Fischer-Tropsch process:

The Fischer–Tropsch process is a series of chemical reactions that convert a mixture of hydrogen and carbon monoxide, also known as syngas, into liquid hydrocarbons. This process is one of the most commonly used techniques for gas-to-liquid (GTL) conversion

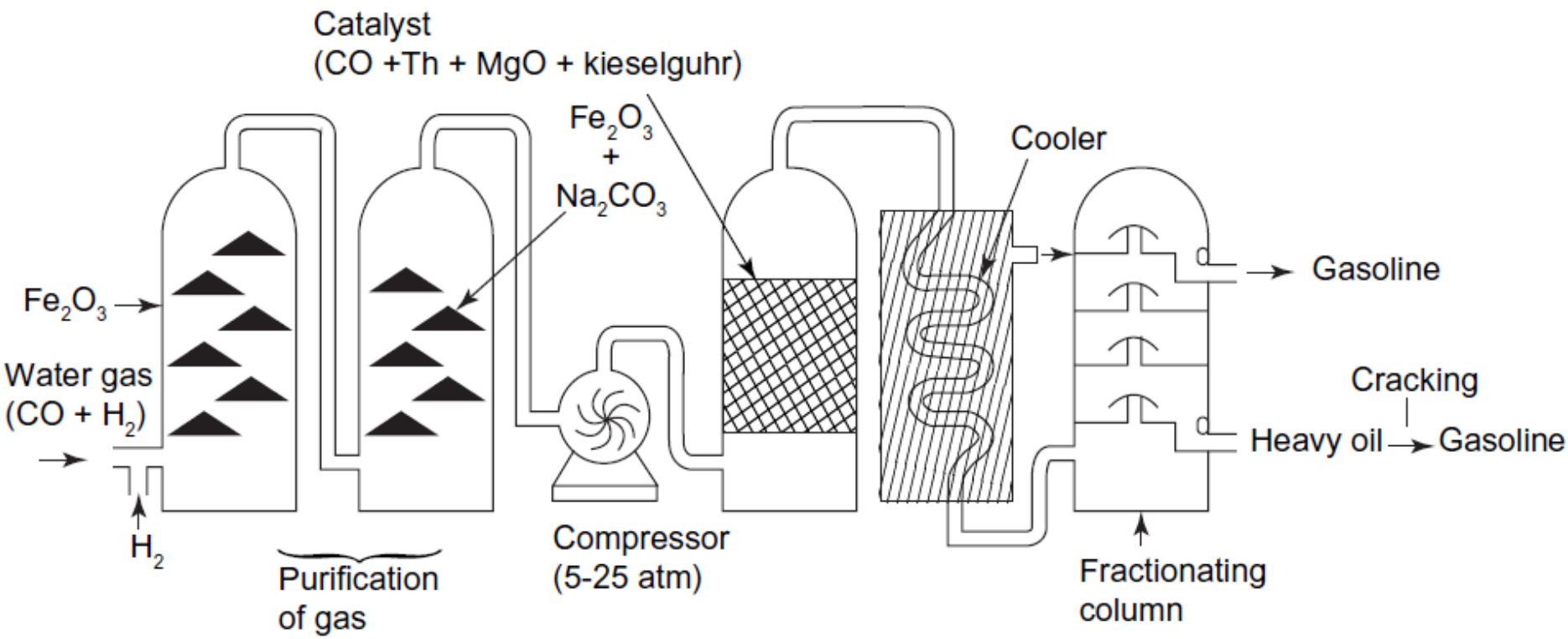
$$C + H_2O \xrightarrow{1200°C} \underset{\text{Watergas}}{CO + H_2}$$

Coke is converted in to mixture of carbon monoxide and hydrogen by passing stream over red hot coke.

$$(2n+1)H_2 + nCO \rightarrow C_nH_{2n+2} + nH_2O$$

$$\underset{\text{Syngas}}{2nH_2 + nCO} \rightarrow C_nH_{2n} + nH_2O$$

The synthesized gas is passed at atmospheric pressure over a cobalt thorium oxide catalyst at 180-200°C to give straight chain paraffins and olefins.



1) Synthesis Gas Formation

$$CH_n + O_2 \xrightarrow{\text{(Catalyst)}} \tfrac{1}{2}\, n\, H_2 + CO$$

2) Fischer-Tropsch Reaction

$$2n\, H_2 + CO \xrightarrow{\text{Catalyst}} -(CH_2-)_n- + H_2O$$

3) Refining

$$-(CH_2-)_n- \xrightarrow{\text{(Catalyst)}} \text{Fuels, lubricants, etc.}$$

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Power Alcohol:

- **Power alcohol is one of the most important non petroleum fuels. These are not used as a prime fuel but used in blends as additives.**

Chemical formula of power alcohol: $C_nH_{2n} + 2OH$

- The first four aliphatic alcohols methanol, ethanol, propanol and butanol can be synthesized chemically or biologically and used as a fuel for internal combustion engines.

Manufacture:

- Methanol can be prepared from bio mass. Power alcohol is manufactured from molasses.

- It is a viscous semisolid material left after crystallization of sugarcane juice. It is a mixture of sucrose, glucose and fructose.

- The molasses are diluted with water to reduce the sugar concentration from about 50-60 percent to 10-12 percent.

- Nutrients like ammonium sulphate, ammonium phosphate and some amount of sulphuric acid is added to maintain the pH value to around 4-5.

- Right proportions of yeast are added and maintained in the temperature of about 30°C.

Thank You